

Auteur: Nathan De Roy
 Studierichting: Informatica
 Oefeningenreeks 3H nr. 12.1

Schets de functie: $r = 3(1 - \sin(\theta))$

$$\left. \begin{array}{l} \text{dom } r = \mathbb{R} \\ P = 2\pi \end{array} \right\} \text{Bep. Dom.} = [0, 2\pi]$$

Nulpunten:

$$r = 0 \Leftrightarrow \sin(\theta) = 0 \Leftrightarrow \theta = \frac{\pi}{2} + k2\pi \text{ met } k \in \mathbb{N} \longrightarrow \theta \in \left\{ \frac{\pi}{2} \right\}$$

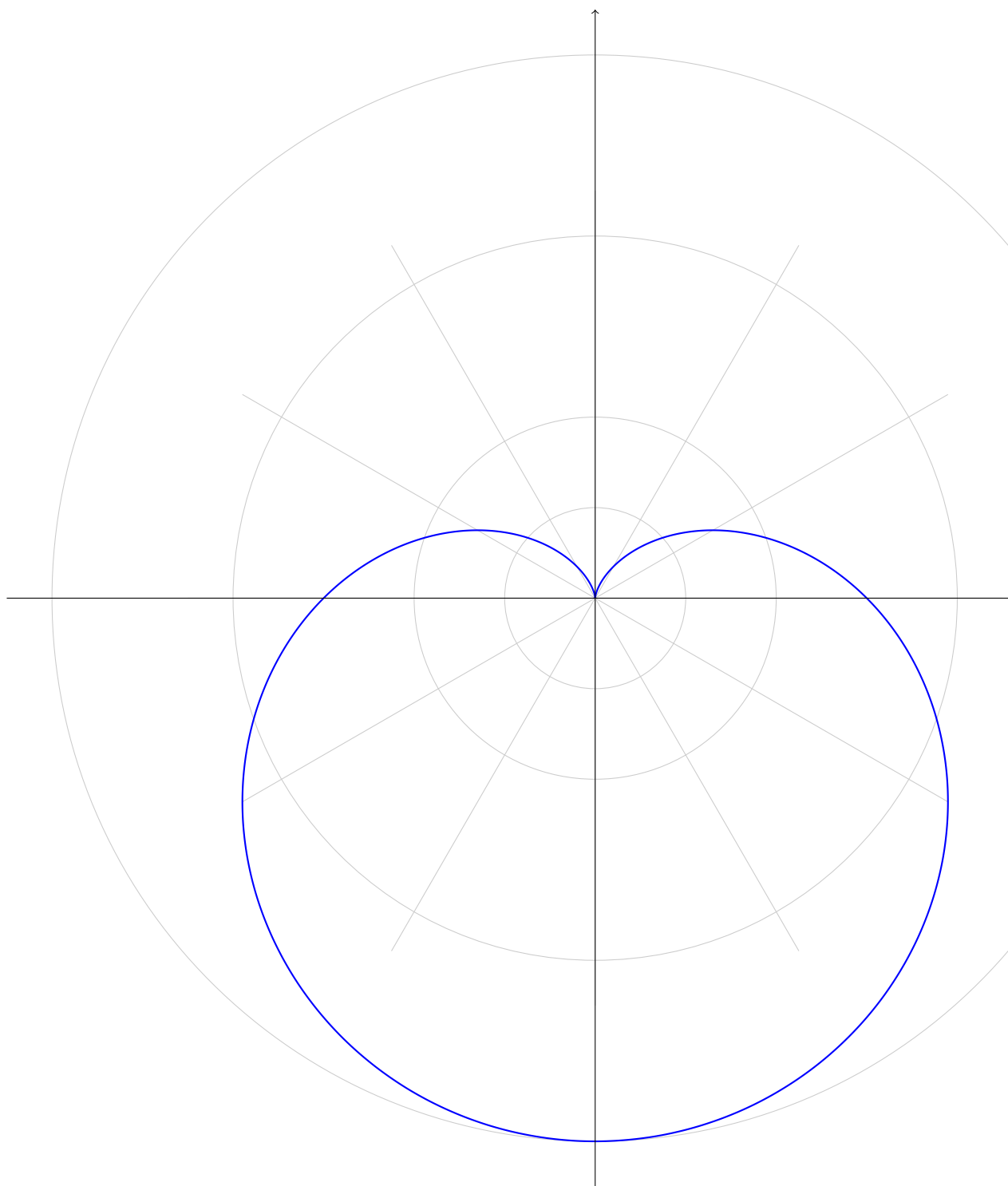
$$r' = 0 \Leftrightarrow -3 \cos(\theta) = 0 \Leftrightarrow \cos(\theta) = 0 \Leftrightarrow \theta = \frac{\pi}{2} + k\pi \text{ met } k \in \mathbb{N}$$

$$\longrightarrow \theta \in \left\{ \frac{\pi}{4}, \frac{3\pi}{4} \right\}$$

Schema:

θ	0		$\frac{\pi}{2}$		π		$\frac{3\pi}{2}$		2π	
r	3	+	0	+	3	+	6	+	3	
r'	-3	-	0	+	3	+	0	-	-3	
	$\searrow$	$\searrow$	min	$\nearrow$	$\nearrow$	$\nearrow$	Max	$\searrow$	$\searrow$	

Grafiek in poolcoördinaten:



References

[1]